

C Programs Samples

1. Write the C program to find the area of circle

```
#include <stdio.h>
void main()
{
    float r , area;
    printf("Enter The Radius of Circle\n");
    scanf("%f", &r);
    area = 3.142* r * r;
    printf("The area of the circle= %f", area);
    return 0;
}
```

2. Write the C program to find the area of rectangle

```
#include <stdio.h>
void main()
{
    float w, h, area;
    printf("enter the height and width\n");
    scanf("%f %f", &h, &w);
    area = h * w;
    printf("Area of the rectangle=%f", area);
    return 0;
}
```

3. Write the C program to find the area of triangle

```
#include <stdio.h>
void main()
{
    float b, h, area;
    printf("enter the height and base \n");
    scanf("%f %f", &h, &b);
    area=(h*b)/2;
    printf("Area of the triangle=%f", area);
    return 0;
}
```

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4. Write the C program to find the area of square

```
#include <stdio.h>
void main()
{
    float side, area;
    printf("enter the side\n");
    scanf("%f", &side);
    area=side*side;
    printf("area of square = %f", area);
    return 0;
}
```

5. Write a C program to calculate Simple Interest

```
#include <stdio.h>
void main()
{
    float p, t, r, si;
    printf("enter the value of Principal, Time and Rate of interest \n");
    scanf("%f %f %f", &p, &t, &r);
    si = (p*t*r)/100;
    printf(" Simple Interest = %f", si);
    return 0;
}
```

6. Write a C program to swap two numbers

```
#include <stdio.h>
void main()
{
    int a, b, temp;
    printf("Enter two number \n ");
    scanf("%d %d", &a, &b);
    temp =a;
    a = b;
    b = temp;
    printf("After swapping a= %d b= %d", a, b);
    return 0;
}
```

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7. Write a C program to swap two numbers without using temporary variable

```
#include <stdio.h>
void main()
{
    int a, b;
    printf("Enter two number \n ");
    scanf("%d %d", &a, &b);
    a=a + b;
    b=a - b;
    a=a - b;
    printf("After swapping a= %d b= %d", a ,b);
    return 0;
}
```

8. Write a c program to convert Fahrenheit to Celsius

```
#include <stdio.h>
void main()
{
    float f, c;
    printf("enter the value in celsius\n");
    scanf("%f", &c);
    celsius=(f - 32.0) * 5.0 / 9.0;
    printf("value in celsius is %f",celsius);
    return 0;
}
```

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9. Write a c program to check whether a number is positive , negative or zero

```
#include <stdio.h>
void main()
{
    int n;
    printf("Enter the number \n ");
    scanf("%d", &n);
    if (n > 0)
        printf("%d is positive.", n);
    else
        if (n < 0)
            printf("%d is negative.", n);
        else
            printf("%d is zero.", n);
    return 0;
}
```

10. Write a c program to check whether a number is even or odd

```
#include <stdio.h>
void main()
{
    int n;
    printf("Enter the number \n ");
    scanf("%d", &n);
    if (n%2 == 0)
        printf("%d is even.", n);
    else
        printf("%d is odd.", n);
    return 0;
}
```

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11. Write a c program to find largest of 3 numbers

```
#include <stdio.h>
void main()
{
    int a, b, c;
    printf("Enter three numbers \n ");
    scanf("%d %d %d", &a, &b, &c);
    if (a > b)
    {
        if(a > c)
            printf("a is larger");
        else
            printf("c is larger");
    }
    else
    {
        if(b > c)
            printf("b is larger");
        else
            printf("c is larger");
    }
    return 0;
}
```

12. Write a c program to check whether a number is prime or not

```
#include <stdio.h>
void main()
{
    int n, i, m=0, flag=0;
    printf("Enter the number to check prime \n");
    scanf("%d", &n);
    for(i=2; i<=n/2; i++)
    {
        if(n % i == 0)
        {
            flag=1;
            break;
        }
    }
}
```

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```
    }  
    }  
    if(flag == 0)  
        printf("Number is prime");  
    else  
        printf("Number is not prime");  
    return 0;  
}
```

13. Write a c program to check for leap year

```
#include <stdio.h>  
void main()  
{  
    int year;  
    printf("Enter the year \n ");  
    scanf("%d", &year);  
    if (((year % 4 == 0) && (year % 100 != 0)) || (year % 400 == 0))  
        printf("%d is a leap year", year);  
    else  
        printf("%d is not a leap year", year);  
    return 0;  
  
}
```

14. Write a C Program to Check Vowel or Consonant

```
#include <stdio.h>  
void main()  
{  
    char ch;  
    printf("enter the character \n");  
    scanf("%c",&ch);  
    if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u' || ch == 'A' || ch ==  
'E' || ch == 'I' || ch == 'O' || ch == 'U')  
        printf("Its a vowel");  
    else  
        printf("Its a consonant");  
    return 0;  
}
```

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15. Write a C program to print n natural numbers using for loop

```
#include <stdio.h>
void main()
{
    int i, n;
    printf("Enter the value of n \n");
    scanf("%d", &n);
    for (i = 1; i <= n; i++)
    {
        printf("%d \t", i);
    }
    return 0;
}
```

16. Write a C program to print n natural numbers using while loop

```
#include <stdio.h>
void main()
{
    int i, n;
    printf("Enter the value of n \n");
    scanf("%d", &n);
    i=1;
    while(i <= n)
    {
        printf("%d \t", i);
        i++;
    }
    return 0;
}
```

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17. Write a C program to find sum of n natural numbers using for loop

```
#include <stdio.h>
void main()
{
    int i, n, sum=0;
    printf("Enter the value of n \n");
    scanf("%d", &n);
    for (i = 1; i <= n; i++)
    {
        sum=sum + i;
    }
    printf("sum = %d", sum);
    return 0;
}
```

18. Write a C program to find sum of n natural numbers using while loop

```
#include <stdio.h>
void main()
{
    int i, n, sum=0;
    printf("Enter the value of n \n");
    scanf("%d", &n);
    i=1;
    while(i <= n)
    {
        sum = sum + i;
        i++;
    }
    printf("sum = %d", sum);
    return 0;
}
```


19. Write a c program to find whether a number is palindrome or not

```
#include <stdio.h>
void main()
{
    int n, rev = 0, rem, org;
    printf("Enter an integer \n ");
    scanf("%d", &n);
    org = n;
    while (n != 0)
    {
        rem = n % 10;
        rev = rev * 10 + rem;
        n = n/10;
    }
    if (org == rev)
        printf("%d is a palindrome.", org);
    else
        printf("%d is not a palindrome.", org);
    return 0;
}
```